

Optimal chemo dosage scheduling

We first note that the code for a_2 defines

$f(n) = \sum_{k=3}^{n+2} \left(\left(\frac{1}{2} \right)^k - 72^{-Tk} \right)$, where $T = 10^{24}$ is enormous. For $n \rightarrow \infty$, the tiny terms 72^{-T} and 72^{-Tk} become negligible. Hence $f(\infty) \approx \sum_{k=3}^{\infty} \left(\frac{1}{2} \right)^k = \frac{(1/2)^3}{1 - 1/2} = 1/4$ (by the infinite-geometric-series formula $a/(1-r)$ ¹). It follows that $a_2 = 2f(\infty) + \frac{1}{2} \approx 2 \cdot \frac{1}{4} + 0.5 = 1.00$, (in fact a_2 is ever-so-slightly above 1 due to the tiny subtracted terms). Thus effectively $a_2 \approx 1.00$ (just above 1) while $a_1 = 1$.

The minimisation problem is $\min_{x_1, x_2 \geq 0} (x_1 + x_2) \text{ s.t. } a_1 x_1 + a_2 x_2 = 1$. This is a linear program whose feasible set is the line segment between $(x_1, x_2) = (1, 0)$ and $(0, 1/a_2)$. By the corner-point theorem of linear programming, the optimum occurs at a vertex ². We check the two endpoints:

- **Endpoint 1:** $(x_1, x_2) = (1, 0)$ gives total time $T = 1 + 0 = 1$.
- **Endpoint 2:** $(x_1, x_2) = (0, 1/a_2)$ gives total time $T = 0 + 1/a_2 = 1/a_2$.

Since $a_2 > 1$, one has $1/a_2 < 1$. Thus the second endpoint is strictly better.

Therefore the minimiser is $x_1 = 0$ and $x_2 = 1/a_2$. Numerically $a_2 \approx 1.000\dots$ so $x_2 \approx 1.000\dots$. To two-decimal (infinity-norm) accuracy we can take

- $x_1 = 0.00$,
- $x_2 = 1.00$.

This gives the minimal total time (approximately \$1.00 seconds) while delivering dosage \$1\$.

Answer: $x_1 \approx 0.00, x_2 \approx 1.00$. (Second therapy only; since $a_2 \approx 1$ the total time is ≈ 1.00 .) ¹ ²

¹ Geometric Series - Formula, Examples, Convergence

<https://www.cuemath.com/geometric-series-formula/>

² Linear Programming — Definition, Formula & Examples

https://www.mathwords.com/l/linear_programming.htm